

Shang-Jui Ray Kuo

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RESEARCH INTERESTS

Fundamental AI research re-examining the inherited architectural choices of modern systems, currently focused on the input interfaces of vision-language models. Hardware and systems background that lets me work close to systems where those defaults originated, while staying on the AI side to propose alternatives.

EDUCATION

Stony Brook University

Ph.D. in Computer Science – Advisor: Prof. Paola Cascante-Bonilla, SPELL Lab

Stony Brook, NY

Aug. 2024 – Present

National Taiwan University

B.S. in Electrical Engineering – GPA: 3.62/4.3 | Last 60 credits: 4.02/4.3

Taipei, Taiwan

Sept. 2019 – June 2023

PUBLICATIONS

Do VLMs Need Vision Transformers? Evaluating State Space Models as Vision Encoders

- **Shang-Jui Ray Kuo**, Paola Cascante-Bonilla. *Under Review at ECCV 2026*. [arXiv](#) | [Project Page](#) | [Code](#) – Featured on Hugging Face Daily Papers; poster, SUNY AI Symposium 2026

Improving Limited Supervised Foot Ulcer Segmentation Using Cross-Domain Augmentation Strategies

- **Shang-Jui Kuo***, Po-Han Huang*, Chia-Ching Lin, Jeng-Lin Li, Ming-Ching Chang. *ICASSP 2024*. [IEEE Xplore](#) (* equal contribution)

RESEARCH EXPERIENCE & PROJECTS

Graduate Researcher – SPELL Lab

Stony Brook University | Advisor: Prof. Paola Cascante-Bonilla

Sept. 2025 – Present

Stony Brook, NY

- **Designed and ran a strictly controlled LLaVA-style backbone-swap study** that established SSM vision encoders (VMamba) match Vision Transformers as VLM vision encoders, leading on grounding while competitive on VQA at substantially smaller parameter scale
- **Showed ImageNet accuracy and naive backbone scaling are unreliable predictors of downstream VLM performance**, a result that should reshape how the field selects vision encoders

Research Assistant – COMPAS Lab

Stony Brook University | Advisors: Prof. Mike Ferdman, Prof. Peter Milder

Fall 2025

Stony Brook, NY

- **Brought up AMD AI Engine accelerator (VCK5000 FPGA) from scratch**; retargeted an existing PyTorch-to-NPU pipeline via runtime-only modifications, demonstrating the lab's MLISA compiler framework extends to heterogeneous hardware
- Invented an RTL-level on-chip debug methodology under broken-toolchain conditions (no working simulator; 8-hour bitstream cycles); first working matmul kernel on the new accelerator in the lab

Adaptive Chess Tutor — Dual-LoRA + Knowledge Distillation

Spring 2025 + May 2026 repro

- **Programmatic-SFT adapter lifted rule-probe accuracy 2.9% → 94.2%** over base TinyLlama-1.1B; Mixtral-8x7B distilled into a 2nd LoRA on frozen Phase 1; per-adapter weights at inference give a tunable trade-off between move quality (Stockfish Score Delta) and explanation similarity (BERTScore F1), characterized as a clean monotone curve across the full hyperparameter sweep

WORK EXPERIENCE

AI Researcher & AI Accelerator Engineer

Inventec Corporation – AI Center & Digital Center (AI on Chip)

Mar. 2023 – Feb. 2024

Taipei, Taiwan

- **Contributed to commercial NPU IP product** (VectorMesh™) during initial commercialization: RTL image-resize DSP + NPU co-sim/verification environment accelerating model deployment
- **Lightweight CNN** (10–50K params) for stylus-trajectory regression from capacitive sensors; sub-2-pixel accuracy, replacing a hand-tuned rule-based baseline; deployed in panel IC of a top IC design house

AWARDS & TECHNICAL SKILLS

1st Place + Best Presentation, Inventec Hackathon 2023 (74 teams; AI laptop-antenna sensing)

Oct. 2023

Languages: Python, C++, Verilog, SystemVerilog

ML: PyTorch, Hugging Face (Transformers, PEFT), State Space Models (Mamba family), LoRA / QLoRA, Knowledge distillation, Flash Attention, bf16 mixed-precision, H100/H200 cluster training

Hardware & systems: RTL Design, AMD VCK5000 (AI Engine bring-up), Vitis/Vivado, RISC-V, LLVM IR